THE DUALIZING FUNCTOR IS AN ANTI-AUTOEQUIVALENCE OF THE BOUNDED COHERENT CATEGORY WITH SQUARE THE IDENTITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement that the internal hom into a dualizing complex preserves the bounded coherent derived category. Invoked at the start of the counted proof (source lines 2832-2834). Shows the functor lands where claimed.
- The evaluation isomorphism identifying an object with the tensor of the double dual of the dualizing complex. Invoked inside the counted proof (source lines 2835-2840) as a cross-chapter pointer. Produces the canonical comparison map to the double dual.

Source reference. Original lemma lemma-finite-ext-into-bounded-injective: sources/stacks-dualizing/pinned-original-source.tex, source lines 2791–2791. Navigation pointer only: the counted text refers to the Stacks item lemma-finite-ext-into-bounded-injective, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-internal-hom-evaluate-isomorphism-technical: sources/stacks-dualizing/pinned-original-source.tex, source lines 2836–2836. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-internal-hom-evaluate-isomorphism-technical, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let A be a Noetherian ring. If $\omega_A^\bullet$ is a dualizing complex, then the functor*

$$D : K \longmapsto R\operatorname{Hom}_A(K, \omega_A^\bullet)$$

is an anti-equivalence $D_{Coh}(A) \to D_{Coh}(A)$ which exchanges $D^+_{Coh}(A)$ and $D^-_{Coh}(A)$ and induces an anti-equivalence $D^b_{Coh}(A) \to D^b_{Coh}(A)$. Moreover $D \circ D$ is isomorphic to the identity functor.

Proof. Let K be an object of $D_{Coh}(A)$. From Lemma lemma-finite-ext-into-bounded-injective we see $R\operatorname{Hom}_A(K, \omega_A^\bullet)$ is an object of $D_{Coh}(A)$. By More on Algebra, Lemma more-algebra-lemma-internal-hom-evaluate-isomorphism-technical and the assumptions on the dualizing complex we obtain a canonical isomorphism

$$K = R\operatorname{Hom}_A(\omega_A^\bullet, \omega_A^\bullet) \otimes^{\mathbf{L}}_A K \longrightarrow R\operatorname{Hom}_A(R\operatorname{Hom}_A(K, \omega_A^\bullet), \omega_A^\bullet)$$

Thus our functor has a quasi-inverse and the proof is complete. □